

Marshfield School History - Instruction Manual

Run guide, setup requirements, editing steps, and database workflows

Contents

- 1) What is needed before running
- 2) How to run the application
- 3) If run fails (quick fixes)
- 4) Project structure and purpose
- 5) Rebuild or refresh the databases
- 6) Editing page text/content
- 7) Hyperlinks
- 8) Table editing
- 9) Layout and style changes
- 10) Staff database page
- 11) Sources database page
- 12) Offline use
- 13) Stop the app

1) What is needed before running

Software

- Windows 10/11
- Python 3.10+ installed (`python --version`)
- Internet only for first-time package install (after that it can run offline)

Important Python note (Windows): Python is not guaranteed to be preinstalled. Check with `python --version`. If it fails or opens Microsoft Store, install Python from <https://www.python.org/downloads/windows/>, tick **Add Python to PATH**, reopen PowerShell, then run `python --version` again.

Project files required

- Full `Marshfield_App` folder, including `app.py`, `requirements.txt`, `templates/`, `static/`, `Images/`, `data/`
- Startup helper files: `start_app.bat`, `install_desktop_shortcut.bat`, `create_desktop_shortcut.ps1`

Important database note: If `data/staff.db` and `data/sources.db` already exist in the project, the application can run without using `init_db.py` or `import_excel_data.py`. Records added through the app are written directly into those existing database files.

2) How to run the application (exact steps)

Open **PowerShell as Administrator** and go to the project:

```
cd C:\path\to\Marshfield_App
```

2. Create and activate virtual environment:

```
python -m venv .venv .venv\Scripts\activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

Create desktop shortcut (run this before starting the server):

```
.\install_desktop_shortcut.bat
```

5. Launch the app by double-clicking the desktop icon **Marshfield School History**.
6. If you do not want to use the desktop icon, you can run manually instead:

```
python -m flask --app app run --debug
```

then open <http://127.0.0.1:5000/>.

Expected startup behavior: / redirects to [/page1](#).

Using **PowerShell as Administrator** from the start helps avoid permission errors while creating the virtual environment, installing packages, or writing to protected folders.

Desktop icon setup (recommended)

Run this once after `pip install -r requirements.txt` and **before** starting the server:

```
install_desktop_shortcut.bat
```

2. This creates a desktop icon called **Marshfield School History**.
3. From next time onward, just double-click that desktop icon to open the app. No manual Python command is needed.

The desktop icon runs `start_app.bat`. That launcher starts the local Flask server on port 5000 (if it is not already running) and opens <http://127.0.0.1:5000/page1> in the browser.

If shortcut creation fails due to permissions or execution policy, run Command Prompt or PowerShell as **Administrator** and run `install_desktop_shortcut.bat` again.

3) If run fails (quick fixes)

- **flask not recognized:** use `python -m flask --app app run --debug`
- **missing package:** run `pip install -r requirements.txt`
- **port in use:** run `python -m flask --app app run --debug --port 5001`
- **permission denied / access denied:** try running PowerShell as Administrator and rerun the command

4) Project structure and purpose

- `app.py`: routes, page rendering, DB search/insert logic
- `templates/base.html`: shared layout
- `templates/pages/*.html`: all page content
- `static/css/styles.css`: colors/layout/style
- `Images/`: linked archive files
- `data/staff.db`, `data/sources.db`: databases
- `start_app.bat`: one-click app launcher
- `install_desktop_shortcut.bat`: one-time desktop icon setup
- `create_desktop_shortcut.ps1`: creates desktop shortcut target

5) Rebuild or refresh the databases

```
python init_db.py python import_excel_data.py
```

If the database files already exist and the app is working normally, you do not need to run these scripts for everyday use or for adding records through the application interface.

6) Editing page text/content

- Edit page text in `templates/pages/*.html`
- Edit shared layout in `templates/base.html`
- Edit styles in `static/css/styles.css`

7) Hyperlinks

Use this format:

```
<a href="/Images/Log%20books/Infant_1877%20-1905.pdf" target="_blank">1877-1905 Infants</a>
```

8) Table editing

1. Open target page template.
2. Copy an existing table row (`<tr>...</tr>`).
3. Update label and link.

9) Layout and style changes

- Main container: `.page`
- Banner image: `.hero`
- Two-column area: `.content-grid`
- Notable dates panel: `aside.panel`

10) Staff database page (Page 5)

- Search by school, dept, last name, position, start year.
- Use + **Add New Staff Record** to insert a record.
- Required field to add: `School Name`.
- New staff records are written directly into `data/staff.db`.

11) Sources database page (Page 6)

- Search by source name, school, source type, department, notes, and hyperlink.
- Use + **Add New Source Record** to insert a record.
- Required field to add: `Source Name`.
- For file links, enter relative path like `Images/...`
- New source records are written directly into `data/sources.db`.

12) Offline use

The app runs offline after initial setup, if Python/packages are already installed and all files in `Images/` are present.

13) Stop the app

Press `Ctrl + C` in the PowerShell window running Flask.

If launched from desktop icon, close the terminal window named `Marshfield App Server` or press `Ctrl + C` in that window.

[Marshfield School History - Instruction Manual](#)